

YERSHOV, Russian Federation

WMO: 341860

Lat: 51.367N

Lon: 48.300E

Elev: 364

StdP: 14.50

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.0	-9.0	-19.7	1.9	-13.7	-14.7	2.5	-8.5	27.1	24.6	24.9	23.3	6.7	20	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	95.0	68.0	91.5	67.1	87.7	65.8	70.3	88.0	68.9	85.8	67.4	83.5	9.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	93.8	74.9	63.3	88.1	74.2	61.6	83.1	73.7	34.4	88.3	33.2	85.9	32.0	83.4	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	20.7	18.5	DB	-19.1	98.8	6.1	3.3	-23.5	101.2	-27.0	103.2	-30.4	105.0	-34.8	107.4
			WB	-19.4	73.0	6.0	2.7	-23.8	74.9	-27.3	76.5	-30.7	78.0	-35.0	79.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	43.9	13.8	14.3	26.0	46.3	60.1	69.1	73.3	70.7	58.4	44.2	30.0	18.6	(6)
(7)		DBStd	23.69	12.16	12.14	9.94	9.91	8.54	7.91	6.85	7.88	8.43	8.43	10.03	11.86	(7)
(8)		HDD50	4885	1124	1000	746	183	16	0	0	0	20	222	601	973	(8)
(9)		HDD65	8438	1589	1420	1210	561	200	46	11	34	232	646	1051	1438	(9)
(10)		CDD50	2653	0	0	1	73	330	573	721	642	271	42	0	0	(10)
(11)		CDD65	731	0	0	0	2	49	169	267	210	34	0	0	0	(11)
(12)		CDH74	7951	0	0	0	31	518	1799	2888	2318	394	3	0	0	(12)
(13)		CDH80	3381	0	0	0	2	134	762	1307	1066	110	0	0	0	(13)
(14)	Wind	WSAvg	9.1	10.1	10.7	10.3	9.9	9.1	8.2	7.7	7.9	8.1	8.7	9.4	9.8	(14)
(15)	Precipitation	PrecAvg	15.4	1.1	0.9	1.0	1.2	1.2	1.7	1.4	1.2	1.7	1.5	1.3	1.2	(15)
(16)		PrecMax	21.7	2.6	1.9	2.2	2.8	2.4	3.9	3.7	3.2	5.7	3.5	4.0	2.4	(16)
(17)		PrecMin	10.4	0.3	0.3	0.0	0.1	0.0	0.1	0.2	0.1	0.1	0.1	0.2	0.4	(17)
(18)		PrecStd	2.9	0.5	0.4	0.5	0.7	0.6	1.0	0.9	0.9	1.5	0.9	0.9	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.5	35.7	56.5	78.1	87.0	97.9	99.6	98.9	88.2	72.4	53.4	41.2	(19)
(20)		MCWB	36.5	33.8	47.6	59.0	64.9	67.6	67.9	68.5	62.7	56.4	49.3	39.4	(20)	
(21)		2%	DB	34.0	34.2	47.3	72.5	83.8	93.1	94.9	94.3	83.1	66.7	48.9	36.6	(21)
(22)		MCWB	33.0	33.0	41.9	55.2	63.4	67.0	68.0	68.3	61.2	53.4	46.1	35.3	(22)	
(23)		5%	DB	32.6	33.1	41.2	67.4	80.3	88.1	91.3	90.6	78.7	61.4	45.7	34.3	(23)
(24)		MCWB	31.9	32.3	37.7	53.4	62.0	65.8	67.4	66.9	59.3	50.9	43.5	33.3	(24)	
(25)		10%	DB	31.2	31.8	37.3	62.4	75.8	83.9	87.6	86.0	73.6	57.0	42.3	32.9	(25)
(26)		MCWB	30.5	31.3	34.8	50.9	59.6	64.9	66.2	65.4	57.6	49.2	40.4	32.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.6	34.5	47.5	61.0	68.7	70.8	72.0	73.9	65.7	58.9	50.9	40.2	(27)
(28)		MDCB	37.1	35.0	54.9	74.2	83.4	89.3	89.7	91.1	82.6	66.4	53.0	40.9	(28)	
(29)		2%	WB	33.3	33.3	42.4	57.3	64.9	68.9	70.5	70.2	63.0	55.1	46.8	35.8	(29)
(30)		MDCB	33.6	33.8	46.4	68.5	80.5	86.1	88.0	88.2	77.4	63.5	48.3	36.3	(30)	
(31)		5%	WB	32.0	32.4	38.2	54.3	62.7	67.4	69.1	68.1	60.9	52.5	43.8	33.6	(31)
(32)		MDCB	32.5	33.0	41.2	65.0	77.2	83.8	85.9	85.9	74.4	59.7	45.3	34.1	(32)	
(33)		10%	WB	30.4	31.1	35.0	51.8	60.6	65.9	67.7	66.3	58.8	50.0	40.6	32.1	(33)
(34)		MDCB	31.2	31.9	36.8	62.2	73.5	81.4	83.3	83.1	71.3	56.0	42.3	32.7	(34)	
(35)	Mean Daily Temperature Range		MDBR	10.5	11.8	12.5	17.7	19.9	20.3	20.4	21.2	19.7	15.1	9.7	9.9	(35)
(36)		5% DB	MCDBR	7.5	8.0	15.7	24.7	24.6	25.5	25.2	25.9	27.3	22.9	12.0	7.2	(36)
(37)			MCWBR	7.5	7.8	11.0	12.2	11.6	9.7	9.1	10.3	12.7	13.3	9.9	7.1	(37)
(38)			MCDBR	7.5	7.9	15.0	22.3	22.0	22.2	22.3	24.0	23.8	20.0	10.6	6.8	(38)
(39)		5% WB	MCWBR	7.5	7.8	10.7	12.0	11.6	9.2	8.8	10.5	12.3	13.0	9.5	7.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.307	0.334	0.372	0.436	0.409	0.394	0.407	0.426	0.376	0.351	0.330	0.306	(40)	
(41)		taud	2.079	2.026	2.064	2.084	2.255	2.350	2.312	2.240	2.350	2.353	2.281	2.140	(41)	
(42)		Ebn at Noon	211	240	256	253	268	273	267	254	255	238	207	191	(42)	
(43)		Edh at Noon	25	36	42	46	41	37	38	39	31	26	21	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	327	676	1088	1437	1872	2035	1985	1685	1154	630	277	194	(44)	
(45)		RadStd	66	86	157	126	133	174	137	124	142	97	41	50	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		+1.30	N/A	N/A	+2.38	N/A	N/A	N/A	N/A	+329	+198	
(47)	Regional (3 neighbors)	N/A	N/A	N/A	+2.79	N/A	N/A	N/A	N/A	+322	+198	(47)

Nomenclature: See separate page